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" Automated Test Framework (ATF) "

Demo & Presented By

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Agenda

Topic 01: Overview

Topic 02: ATF : Architecture & core objects

Topic 03: ATF : Application Modules

Topic 04: Roles in ATF

Topic 05: Runners in ATF

Topic 06: Schedules in ATF

Topic 07:What can be automated in ATF?

Topic 08: ATF Limitation & Challenges

Topic 09: Maintenance of ATF

Topic 10: How ATF Helps in Platform Upgrades?

Topic 11:Demo Use Case

Topic 12:Pro Tips/Hacks in ATF

Final Q&A Session

Overview:

What is ATF?

- Automated Test Framework, a tool that allows you to create and run automated tests on your instance to verify applications, configurations, and business logic without manual intervention.

Why We Need ATF?

- ATF helps reduce development and upgrade time by replacing repetitive manual testing with reusable, automated test cases that can be run after modifications or during upgrades to ensure the platform functions as expected.

ATF Key benefits

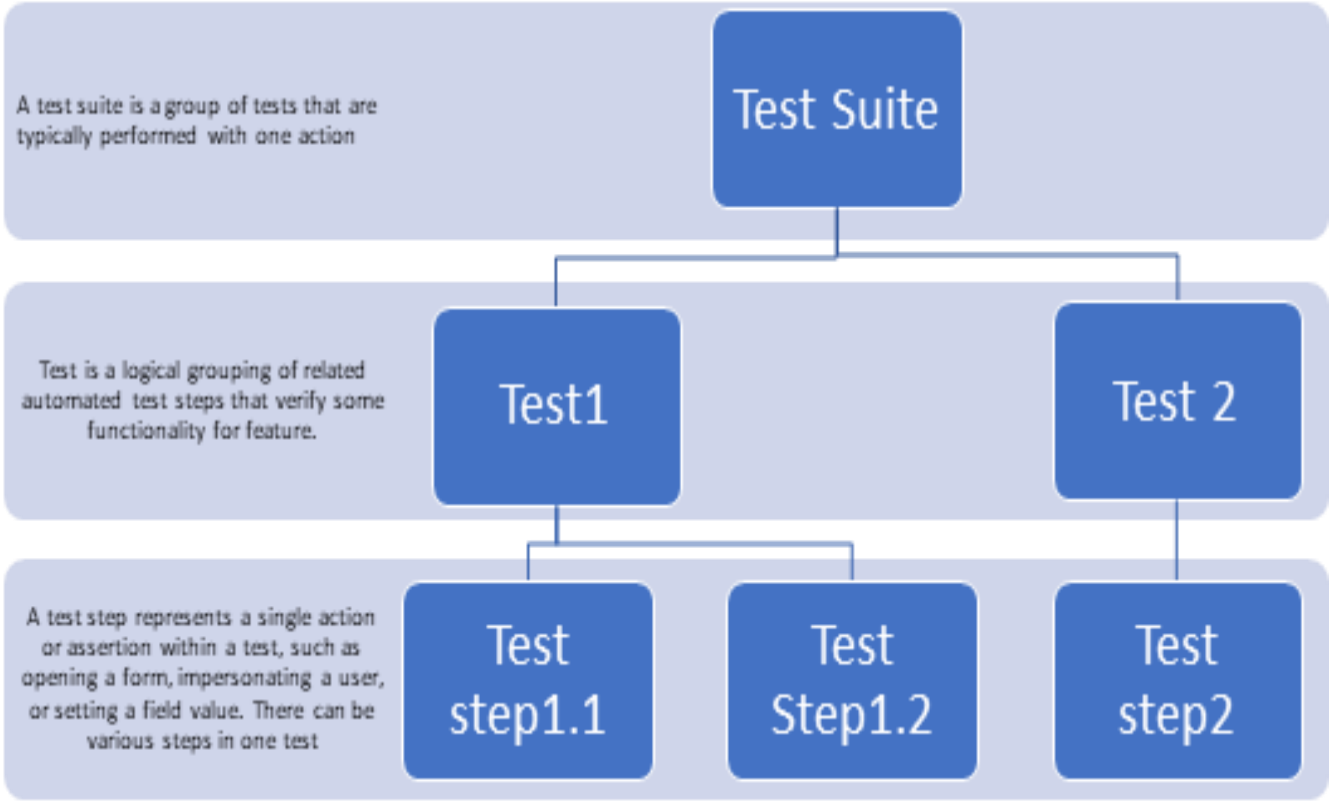
- Faster, repeatable regression testing
- Confident upgrades and deployments
- Reusable test steps and suites
- Automatic cleanup / rollback of test data

Business need / Problem statement

- Manual regression is slow, inconsistent and expensive
- Upgrades and frequent changes risk regressions in critical workflows
- We need repeatable, fast validation for releases

ATF : Architecture & core objects

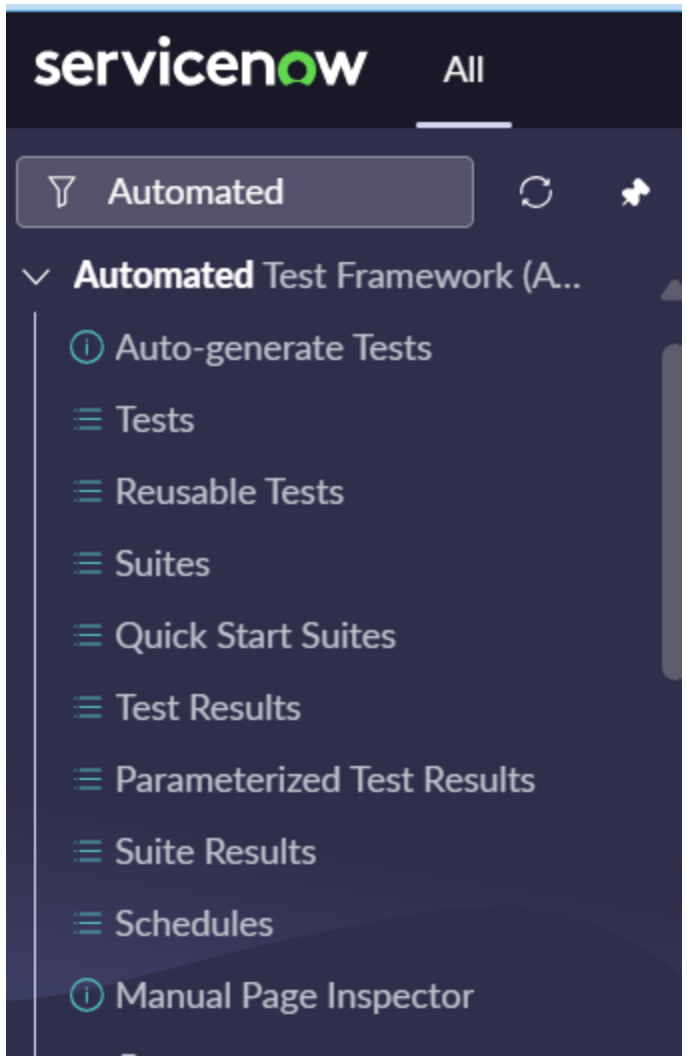
- **Test Step** — single action or assertion
- **Test Case (Test)** — ordered set of steps (one scenario)
- **Test Suite** — group of test cases (e.g., upgrade regression)
- **Test Runner / Results** — execution, screenshots, logs

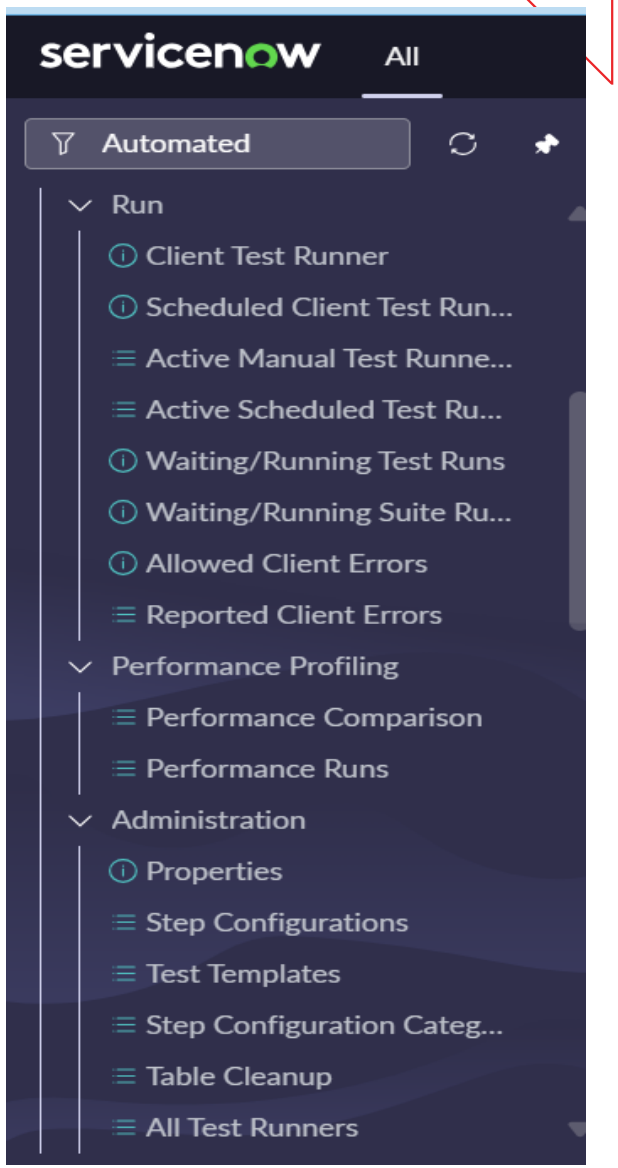


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ATF : Application Modules

- Tests
- Suites
- Test Results
- Suite Results
- Schedules
- Parameterized Test Results
- Step Configurations
- ATF Properties / Settings
- Test Templates





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Roles in ATF:

🛠️ atf_test_designer	⚙️ atf_test_admin	🌐 atf_ws_tester	🔑 admin (System Administrator)
• Create/modify test cases & steps, run tests, validate results, update tests after changes	• Manage ATF settings, reusable step library, test suites, schedules, and governance standards	• Build & run API/web service tests (REST/SOAP), verify responses and integrations	• Enable ATF plugin, assign roles, maintain instance/data cleanup, manage permissions & security

Runners in ATF:

Client Test Runner

- Performing other activities while tests are being executed can result in those activities being performed by the currently executing user and could be rolled back.
- Opens a real browser window and runs the test step-by-step — like a virtual user clicking through the app.

Cloud Runner

- Performing other activities while tests are being executed can be possible
- Runs tests in the background on ServiceNow's cloud infrastructure — no browser window opens, faster & scalable.

Prerequisites for Running Cloud Runner:

✅ **Need to Install ATF Test Generator and Cloud Runner in Plugins Store**

✅ **Need to Create Service Account (Name- ATF Cloud User) in User Table with help of operation team and select as Servicenow service account cloud user**

Schedules in ATF:

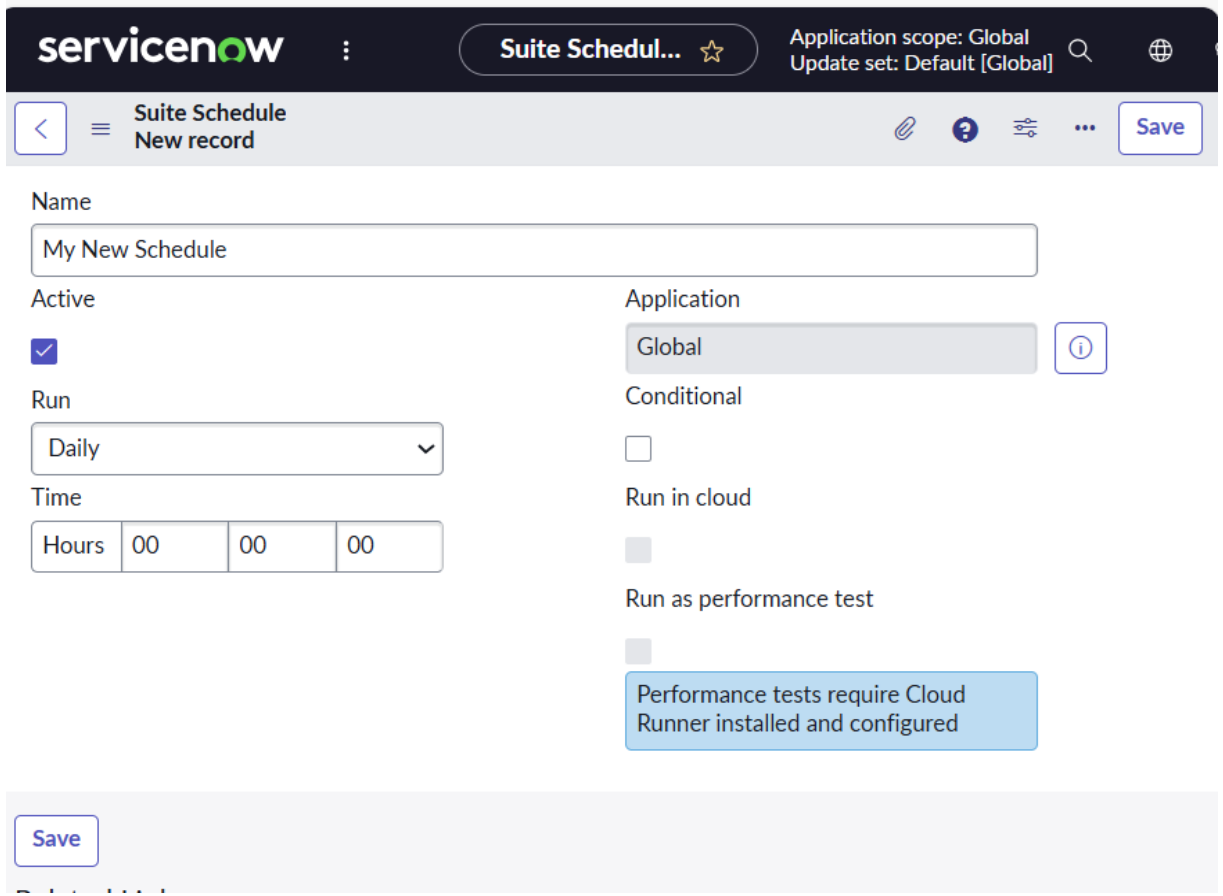
•What Are Schedules?

Schedules in ATF are like alarm clocks for your automated tests — Suites, they tell ServiceNow when to run a set of test Suites automatically

•Why Schedules Matter:

*Catch issues early – tests can run overnight or after deployments.
*Save time – no manual intervention needed.
*Consistency – same tests run the same way, every time

Prerequisites:
For Running Schedules We Need Installation of Cloud Runner



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What Can Be Automated in ATF (ServiceNow?

- ✅ **Forms & Fields** – Open forms, fill fields, submit records
- ✅ **Workflows** – Validate business rules, approvals, notifications
- ✅ **UI Actions** – Test buttons, links, UI policies, client scripts
- ✅ **List Operations** – Search, filter, update list records
- ✅ **Email & Notifications** – Check if notifications are triggered
- ✅ **Integrations** – REST/SOAP API calls and response validation
- ✅ **Regression Tests** – Run after upgrades or deployments
- ✅ **Data Setup & Cleanup** – Prepare test data automatically

ATF : Limitations / Challenges

Being realistic builds credibility.

- Tests need maintenance: when UI or workflows change, some tests break. So someone must update.
- Some things ATF is *not* good at: load/performance testing, very graphical UI / animation behavior, unpredictable third-party integrations.
- Initial investment: writing tests, setting up best practices, training the team, might slow initial pace.
- Environment requirement: should run in non-prod; data setup / cleanup; sufficient access.
- False positives/negatives: test logic needs to be precise.
- Over-automation risk: automating everything may not be efficient; need to choose what's worth automating

Maintenance of ATF:

- ✅ **Update Sets** – Create current update sets and include ATF developments and export to XML once done and save it on SharePoint.
- ✅ **Keep Tests Modular & Focused** – one scenario per test, combine with suites for coverage
- ✅ **Use Naming Standards** – Use consistent names for tests, steps, and suites.
- ✅ **Customization** – Always update tests when new requirements arise and re-run tests.
- ✅ **Schedules** – Ensure schedules are running and review scheduled results regularly.
- ✅ **Failed Tests** – Investigate and fix intermittent failures promptly — flaky tests reduce trust in automation.
- ✅ **Reports** – Publish ATF results dashboards or reports (or maintain manual Excel sheets if needed).
- ✅ **Assign Ownership** – Have an ATF champion or QA owner maintain tests and the shared library.

✅ **Use Dynamic Data & Rollback** – Avoid hardcoding sys_id; use rollback to clean up after test execution

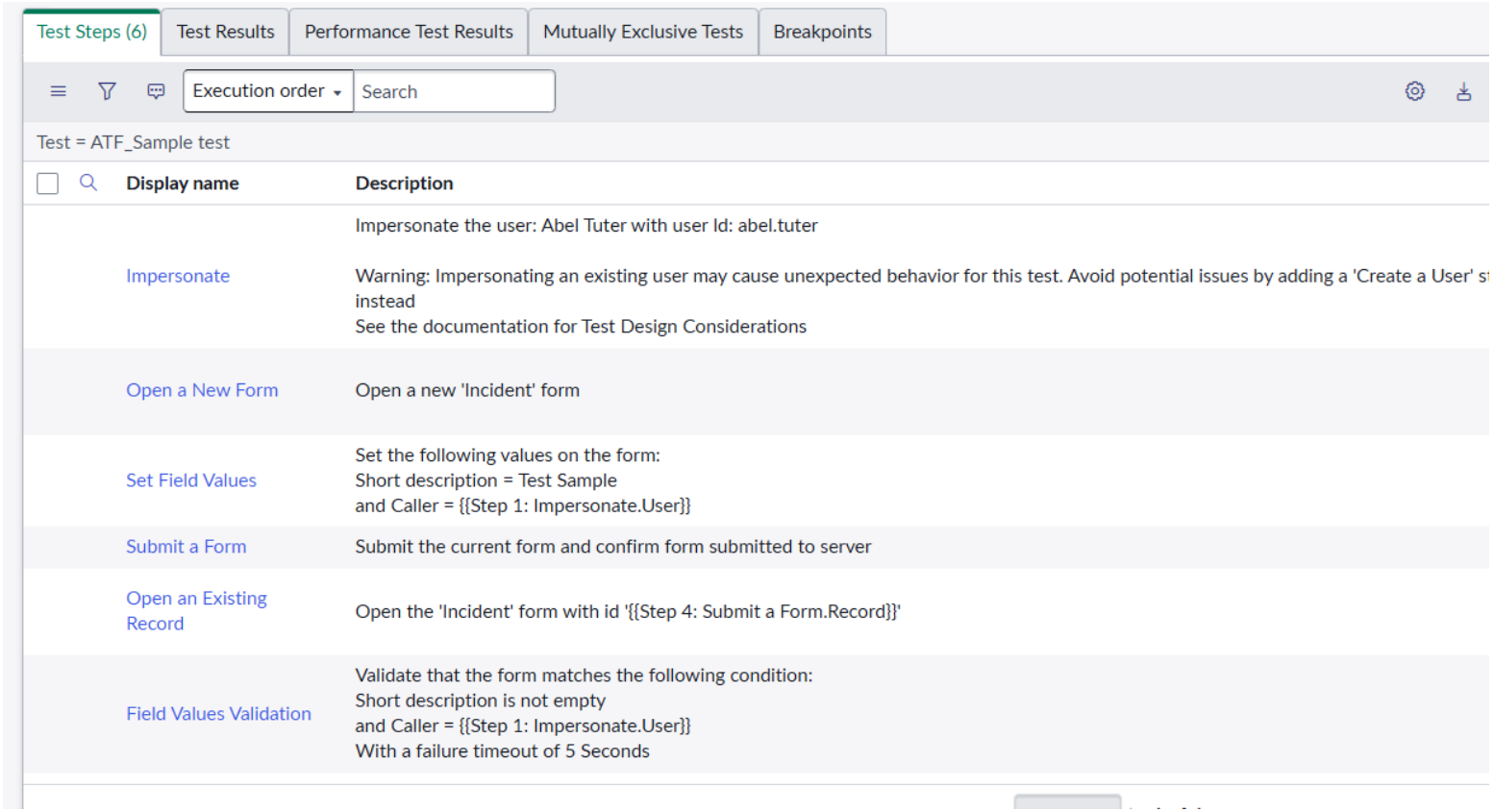
How ATF Helps in Platform Upgrades?

- **Automates Regression Testing-** Runs prebuilt and custom tests automatically after the upgrade and Verifies critical business processes without manual effort
- **Early Issue Detection**- Identifies broken functionality or missing customizations quickly and Helps fix issues before users are impacted
- **Speeds Up Upgrade Cycles**- Reduces the time spent on manual testing and Enables faster go/no-go decisions for production deployment
- **Ensures Consistency**- Provides repeatable, reliable test results across environments
- **Reduces Risk**- Improves confidence that the instance works as expected post-upgrade and

Demo use Case

➢ Incident Management Test & Steps

- ❑ Impersonate
- ❑ Open a New Form
- ❑ Set Field Values
- ❑ Submit a Form
- ❑ Open an Existing Record
- ❑ Field Values Validation



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Pro Tips in ATF

✅ **SN Utils Extension**- Add SN utils Extension in your browser for identifying technical names in forms easily

✅ **Dynamic Approver** – Use *GlideRecord* to fetch approvers dynamically instead of hardcoding a specific user.

✅ **Date Selection** – Use `javascript:gs.daysAgo(0)` instead of manually selecting a date from the calendar.

✅ **Cloud Runner** – Use the 2 free Cloud Runners provided by ServiceNow for faster execution; consider purchasing additional runners for large regression suites.

✅ **Attachment** – When selecting an attachment field in a form, use the corresponding `sys_id` from the Attachment table.

✅ **Server Error Handling** – For server errors in failed tests, click **"Add errors to warning list"** on the test results page or uncheck the **"Fail on server error"** checkbox on the test page if appropriate.

✅ **Parameterized Testing** – Use Parameterized Testing to handle large sets of data with the same workflow in a single test case.

Thank you

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